

Nova Electronics Corp — Marketing Value Creator (MVC)

System Architecture, C4 Component Models & Design Specifications

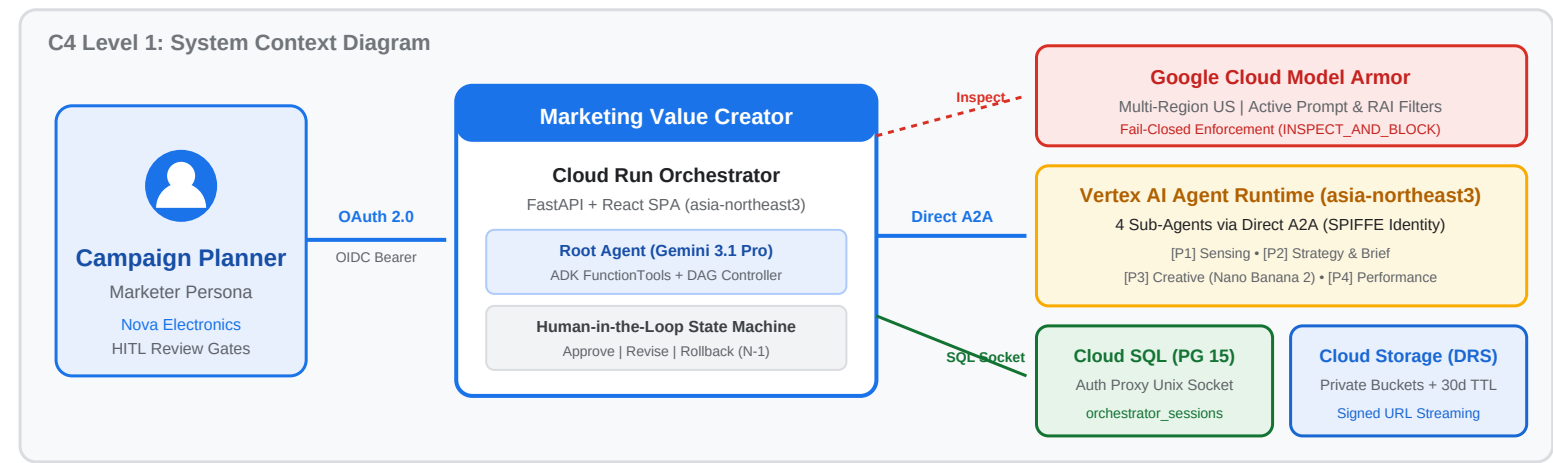
Forward Deployed Engineer: Ryan Ahn (ryanahn@) • Capstone Project • September 2026

Executive Architecture Summary: Marketing Value Creator (MVC) is an enterprise generative AI campaign planning and simulation platform designed to compress Nova Electronics Corp's manual 4-to-6-week multi-agency briefing workflow into an interactive, multi-agent simulation taking under 15 seconds of compute. The platform operates under strict brand and security guardrails with 100% mathematical budget conservation and zero arithmetic hallucination.

Architectural Dimension	Technical Design Implementation	Business & Operational Benefit
Topology & Deployment	FastAPI Orchestrator on Cloud Run (asia-northeast3) serving React 19 SPA static assets under a unified origin with Direct VPC Egress and Zero-Trust Firewall (Default Deny + Whitelist).	Zero CORS complexity, sub-50ms regional latency in Seoul, and automatic scale-to-zero cost efficiency.
Multi-Agent Orchestration	4 specialized sub-agents on Vertex AI Agent Runtime orchestrated via Direct A2A JSON-RPC protocol (SPIFFE identity) with ADK Root Agent FunctionTools.	Modular domain separation ([P1] Sensing, [P2] Strategy, [P3] Creative, [P4] Performance) with zero context dilution.
Asymmetric Model Tiering	Gemini 3.5 Flash Lite for subagents (P1, P2, P4); Nano Banana 2 Lite (gemini-3.1-flash-lite-image) for P3 visuals; Gemini 3.1 Pro for root orchestration and evaluation.	FinOps optimization: \$0.0455 / run actual vs \$0.10 target ceiling (54.5% below target).
Security & Guardrails	Google Cloud Model Armor (version1-guardrails in multi-region US) with active prompt/RAI/SDP filters, fail-closed policy, Google OAuth 2.0 OIDC, and private Cloud SQL.	Zero public DB exposure, 100% prompt injection defense, and enterprise brand safety compliance.
Quality Flywheel & CI/CD	Cloud Build Automated Dual Gate (119 passing tests: 102 unit + 17 integration + pre-prod eval gate on 9 golden scenarios with LLM Judge >= 4.0/5.0).	Blocks release regressions automatically; enforces 100.0% budget conservation and 100% schema validity.

1. System Context Diagram (C4 Level 1)

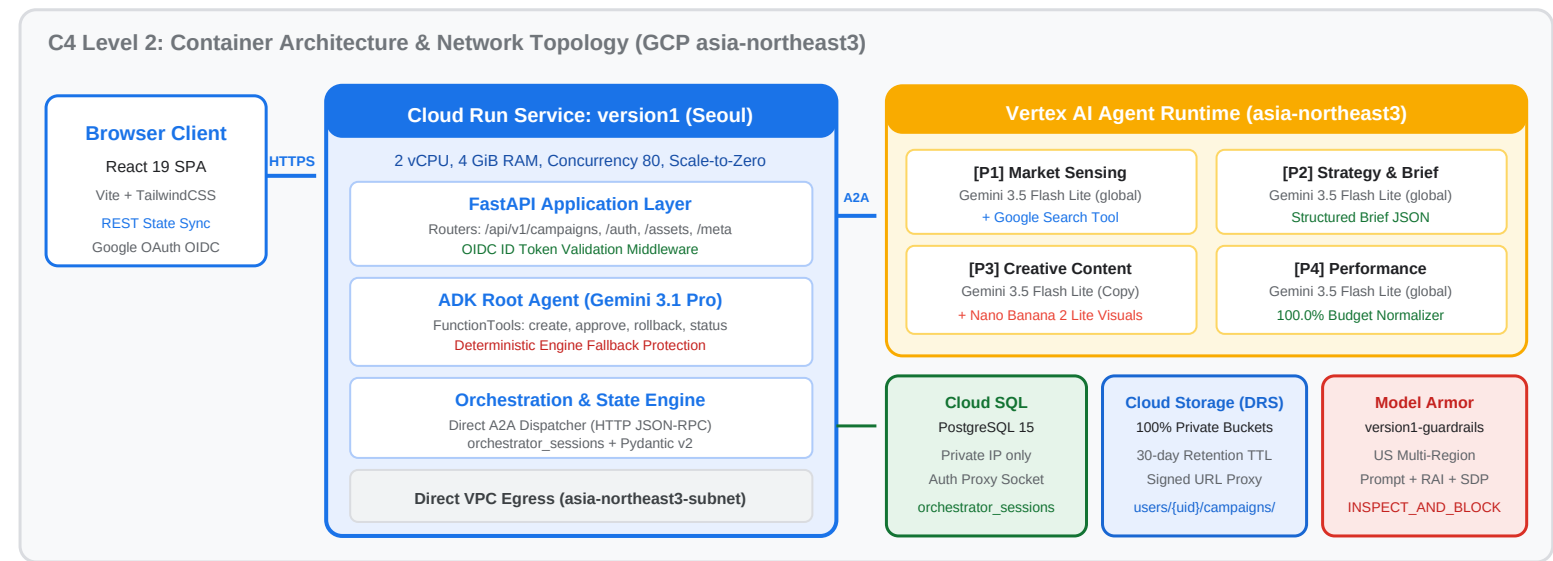
Maps the overarching boundary of the MVC system, user personas (Marketer / Campaign Planner), and external cloud dependencies.



Boundary Component	Protocol / Integration	Security & Trust Boundary	Data Classification & Retention
Campaign Planner (User)	HTTPS Browser / React 19 SPA	Google OAuth 2.0 OIDC Bearer Token verification on Cloud Run endpoint.	Marketer email and role (Viewer / Editor / Admin).
MVC Orchestrator	FastAPI on Cloud Run (asia-northeast3)	Direct VPC Egress; IAM Workload Identity; isolated container runtime.	Session state & intermediate stage deliverables (30-day TTL).
Vertex AI Agent Runtime	Direct A2A JSON-RPC Protocol	SPIFFE-based Agent Identity; authenticated internal RPC	Domain deliverables: market sensing, brief, creative, insights.
Google Cloud Model Armor	REST API (US Multi-Region)	Fail-closed policy (INSPECT_AND_BLOCK); pre-execution prompt & RAI inspection.	Transient inspection logs; security violation audit trails.

2. Container Architecture & Network Topology (C4 Level 2)

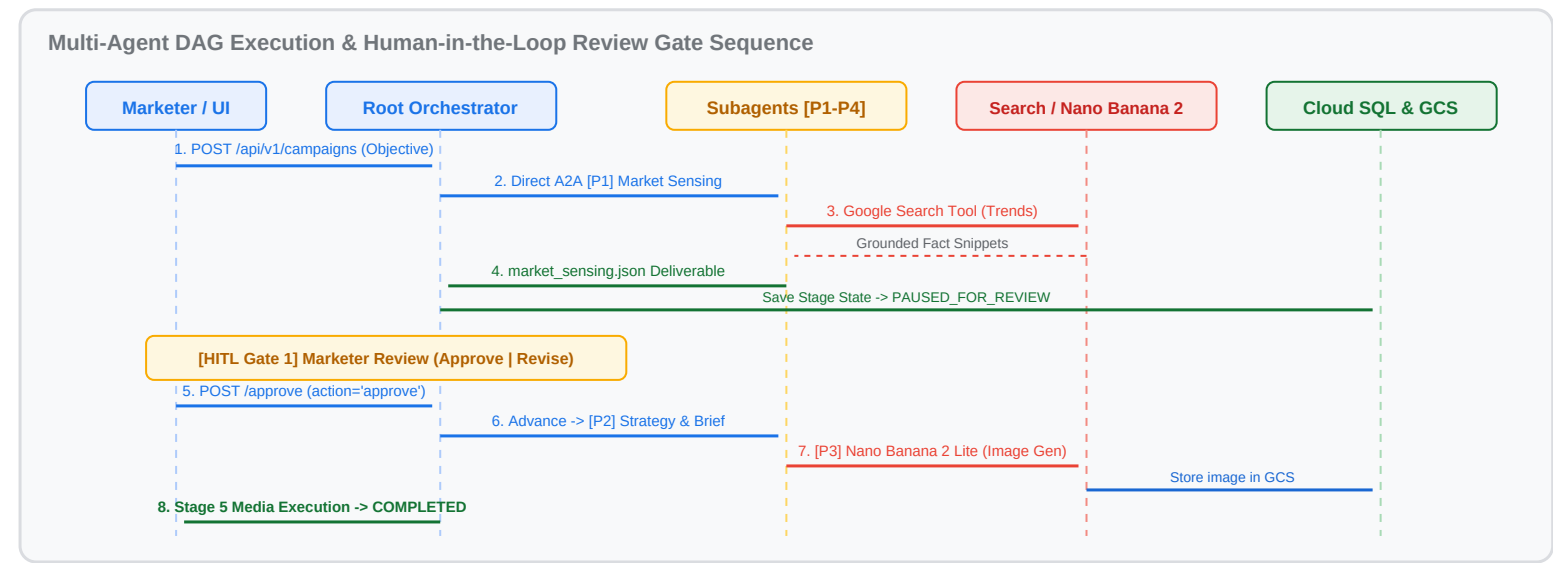
Detailed layout of Cloud Run, Direct VPC Egress, Cloud SQL Auth Proxy, and external Vertex AI model services.



Container / Resource	Sizing & Runtime Specs	Network & Security Controls	Design Rationale
Cloud Run Service	2 vCPU, 4 GiB RAM, concurrency 80, min 0, max 10	Direct VPC Egress (asia-northeast3-subnet 10.10.0.0/24); Zero-Trust Firewall (Default-Deny Ingress/Egress, Whitelist: 443, 3307, 53)	Eliminates VPC connector bottleneck; enables private IP database routing.
Cloud SQL PostgreSQL 15	db-f1-micro (shared vCPU, 0.6 GiB RAM)	Private IP only; Auth Proxy Unix domain socket at /cloudsql	Zero public IP attack surface; durable session persistence in orchestrator_sessions.
Vertex AI Foundation Models	Gemini 3.5 Flash Lite + Gemini 3.1 Pro + Nano Banana 2 Lite	Endpoints explicitly pinned to location='global'	Accesses worldwide GPU/TPU pool, avoiding regional model quota and 404 errors.

3. Multi-Agent DAG & Human-in-the-Loop Review Sequence

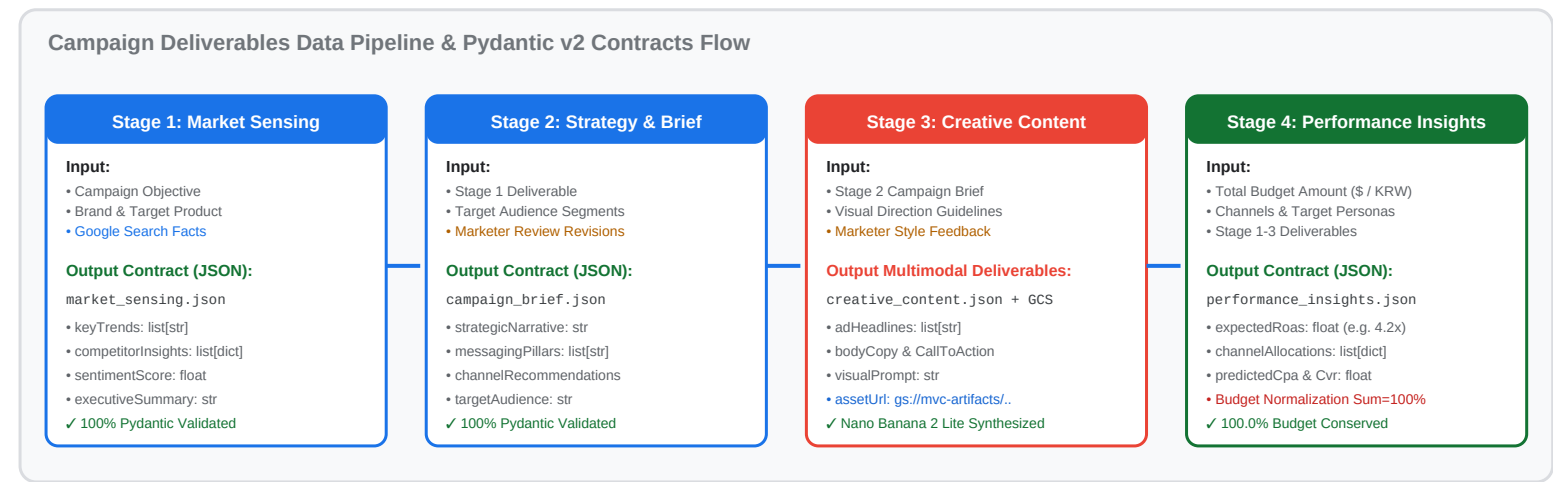
Illustrates sequential subagent execution across P1-P4 and the asynchronous Human-in-the-Loop review gates.



Execution Stage	Agent & Tool Assigned	Human Review Interaction (HITL)	Invariance / SLA Constraint
Stage 1: Market Sensing	Gemini 3.5 Flash Lite + Google Search Grounding Tool	Marketer inspects competitor findings and trend radar.	Turn latency < 3.0s; live grounded market facts.
Stage 2: Strategy & Brief	Gemini 3.5 Flash Lite (Structured Synthesis)	Marketer approves strategic narrative or provides revision feedback.	100% Pydantic validation on campaign_brief.json
Stage 3: Creative Content	Gemini 3.5 Flash Lite (Copy) + Nano Banana 2 Lite (gemini-3.1-flash-lite-image)	Marketer reviews copywriting and visual mockup image.	Visual generation latency < 8.0s; asset stored in DRS GCS.
Stage 4: Performance Insights	Gemini 3.5 Flash Lite + Deterministic Budget Normalizer	Marketer inspects ROAS projections and channel allocation.	100.0% Budget Conservation (Sum of channels == Total Budget).

4. Deliverables Pipeline & Data Contract Architecture

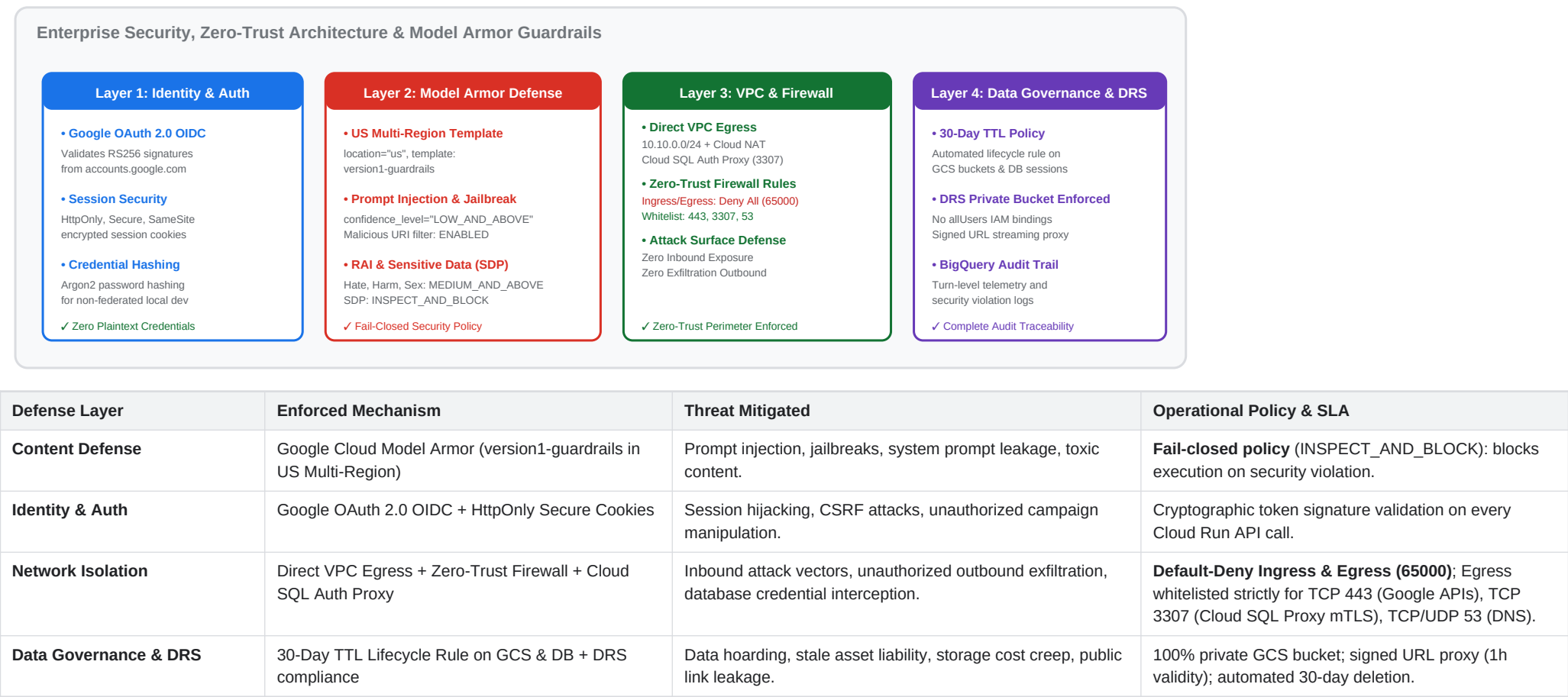
Contract-first data flow showing Pydantic v2 schemas passed between stages without context dilution.



Contract Schema	Pydantic v2 Model Path	Key Fields & Data Types	Validation & Error Handling
MarketSensingDeliverable	app.schemas.deliverables	keyTrends (list[str]), competitorInsights (list[dict]), sentimentScore (float)	Strict JSON mode enforcement; fallback to prompt objective on parse failure.
CampaignBriefDeliverable	app.schemas.deliverables	strategicNarrative (str), messagingPillars (list[str]), targetAudience (str)	Incorporates marketer revision feedback from stage history snapshots.
CreativeContentDeliverable	app.schemas.deliverables	adHeadlines (list[str]), visualPrompt (str), assetUrl (str)	Generates high-res image via Nano Banana 2 Lite (gemini-3.1-flash-lite-image); fallback to SVG placeholder.
PerformanceInsightsDeliverable	app.schemas.deliverables	expectedRoas (float), channelAllocations (list[ChannelAlloc]), predictedCpa	Programmatic normalization ensures channel percentages sum exactly to 100.0%.

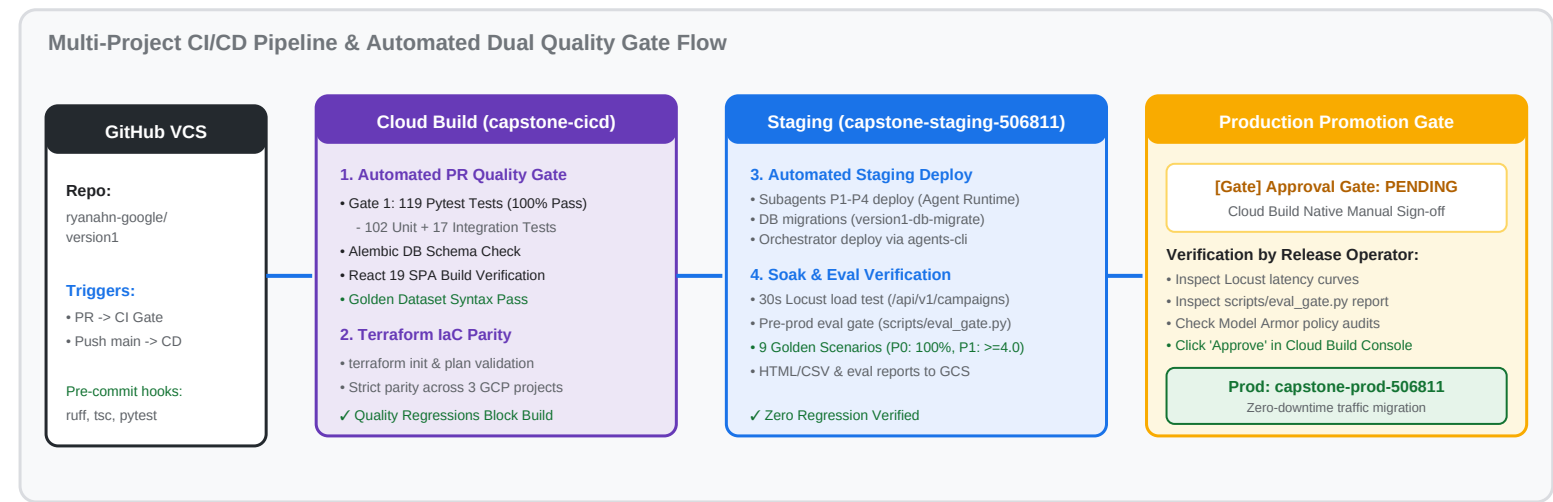
5. Security, Zero-Trust Architecture & Model Armor Perimeter

Four-tier defense-in-depth model encompassing identity, content guardrails, network isolation, and data governance.



6. Multi-Project CI/CD Pipeline & Automated Dual Quality Gate

Google Cloud Build pipeline enforcing 100% test passing, automated Golden Eval benchmark, and headless Locust load verification.



Pipeline Stage	Automated Action	Quality Gate Criterion	Promotion / Blocking Behavior
Pre-Commit Hook	Local developer git commit validation script	ruff format, ruff check, tsc --noEmit, pytest (119 tests)	Fails immediately if any check, linter, or test fails locally.
Pull Request (Gate 1)	Cloud Build trigger: pr-version1	119 pytest tests (102 unit + 17 int) + alembic check + npm build	100% pass required; merges to main are strictly blocked on failure.
Pre-Prod Eval (Gate 2)	scripts/eval_gate.py on 9 golden scenarios	P0: 100% Schema & Budget, 0 Failures; P1: Judge Score >= 4.0/5.0, Regression <= 0.2	Blocks release if prompt regression, schema drift, or budget math fails.
Staging Load Test	30s headless Locust verification against staging Cloud Run	0% HTTP error rate; P95 turn latency within SLA thresholds	Generates HTML/CSV audit reports; triggers production approval gate.
Production Promotion	Cloud Build trigger: deploy-version1	approval_config { approval_required = true }	Enters PENDING state until release operator manually approves in console.

7. Cloud SQL Relational Data Model & Session Persistence Schema

PostgreSQL 15 relational schema storing user credentials, campaign sessions, intermediate deliverables, and rollback snapshots.

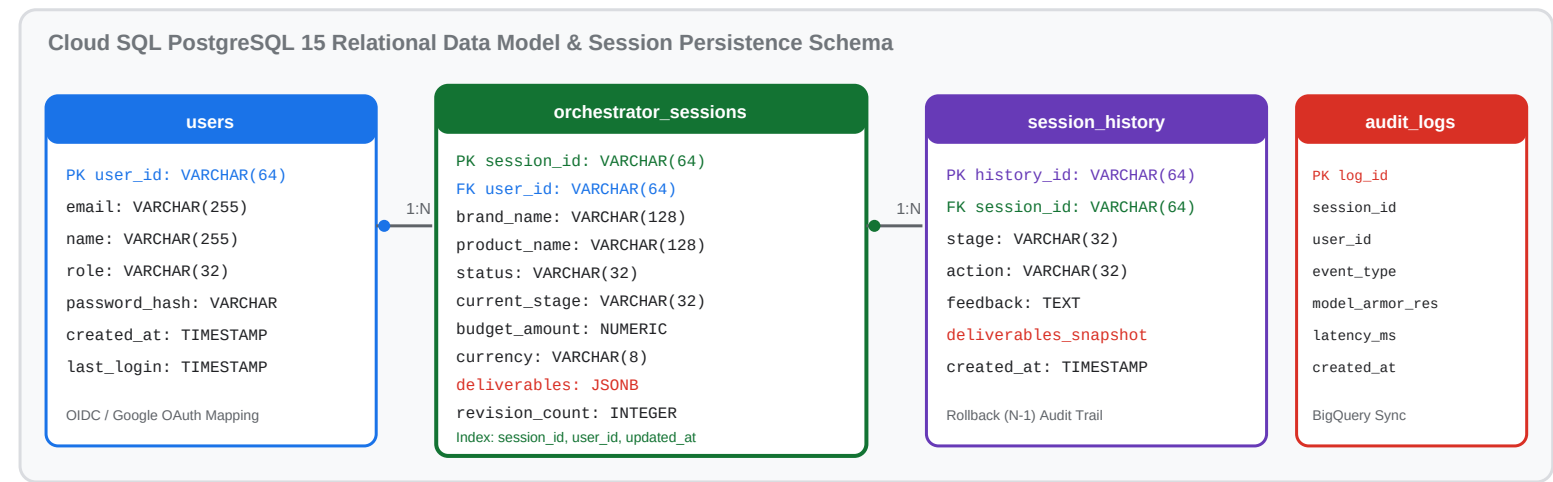


Table Name	Primary Key / Foreign Keys	Core Columns & Data Types	Indexing & Optimization
users	PK: <code>user_id</code> (VARCHAR 64)	<code>email</code> (VARCHAR 255), <code>name</code> , <code>role</code> (VARCHAR 32), <code>password_hash</code> , <code>created_at</code>	Unique index on <code>email</code> ; fast lookups during OAuth login.
campaign_sessions / orchestrator_sessions	PK: <code>session_id</code> FK: <code>user_id</code> -> <code>users.user_id</code>	<code>brand_name</code> , <code>product_name</code> , <code>status</code> , <code>current_stage</code> , <code>budget_amount</code> , deliverables (JSONB)	B-tree index on (<code>user_id</code> , <code>updated_at</code> DESC) for dashboard queries.
session_history	PK: <code>history_id</code> FK: <code>session_id</code> -> <code>campaign_sessions</code>	<code>stage</code> , <code>action</code> ('approve' 'revise' 'rollback'), <code>feedback</code> , deliverables_snapshot (JSONB)	Enables instant N-1 rollback without re-running previous subagents.
audit_logs	PK: <code>log_id</code>	<code>session_id</code> , <code>user_id</code> , <code>event_type</code> , <code>model_armor_result</code> , <code>latency_ms</code> , <code>created_at</code>	Append-only log; streamed to BigQuery Agent Analytics for fleet monitoring.